Report – QIST Project:
Cooling Algorithms for Quantum State Preparation

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Abstract

We numerically study a bath-assisted quantum cooling channel for finite transverse field Ising models with $N = 4$–6 spins. The channel is implemented by simulating the system–bath evolution with a two-level thermal ancilla, followed by ancilla reset, and is averaged over local Pauli coupling operators and bath frequencies as suggested in Ding et al. [2026]. Using exact diagonalization, we compute the channel fixed point and compare it with the exact Gibbs state. While applicability of the cooling channel beyond weak-coupling is treated in Wang and Ding [2025] our observations for the chosen model also go along these lines and extend the weak-coupling analysis into a finite-size numerical regime, while remaining subject to discretization, frequency-quadrature, and finite-size effects.

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Introduction

Measurements of observables $\langle \mathcal{O} \rangle$ only make sense once the state of the system, either pure or mixed, is fixed. A well-known result from statistical mechanics is that for systems that are in thermal equilibrium in the canonical ensemble, the states under consideration are the Gibbs states ρ_β in the case of finite temperature or the ground-state $|\Psi_0\rangle$ in the case of zero-temperature. Thus preparing thermal and ground states is of considerable interest and many approaches to these ends have been considered.

The preparation of these states is a nontrivial task and in full generality as hard as it can get: for general Hamiltonians ground-state preparation is known to be a QMA-complete problem, which, loosely speaking is the quantum analog of an NP-hard problem. Also thermal state preparation is known to be NP-hard in some cases. Therefore efficient preparation algorithms that work for important classes of Hamiltonians are of great interest.

One particularly important example of expectation values with respect to thermal states is the time-ordered Green's function of a many-body Hamiltonian H at a given inverse temperature β: it is the expectation value of the *correlation function* $-i\langle \mathcal{T} c_i(t) c_j^\dagger(t') \rangle_\beta$. Once it is known, several important one-particle properties can be effectively computed, e.g. spectral function, density of states, particle density, self-energy etc. It is therefore essential to have reasonably accurate estimates of these thermal states resp. the ground state available, since due to dimensional blow-up exact computations are infeasible. There are several routes: Among the most well-known are variational methods such as matrix product states, imaginary time evolution and adiabatic

evolution. Another approach is to engineer a specific interaction with a bath, that effectively enforces thermalization at a given temperature. Such a dissipative system not only has the advantage of being implementable on a quantum computer or quantum simulator, but also allows for running times that for certain relevant systems are often merely polynomial in the system's parameters. This route of using dissipation as an effective way to obtain the desired states is the object of the present work.

We numerically investigate a small model system and use a dissipative algorithm that has been introduced in Ding et al. [2026]. Here the total system's Hamiltonian has three contributions

$$H_{\text{cooling}} = H_0 + H_{\text{int}} + H_{\text{bath}}, \tag{1}$$

where H_0 denotes the Hamiltonian of the system whose thermal states should be approximated, H_{bath} the Hamiltonian of the environment and H_{int} the dissipative action between the two. A cooling protocol would typically look like

$$\rho \otimes \rho_b \longrightarrow U(t)(\rho \otimes \rho_b)\, U^\dagger(t) = \tilde{\rho} \longrightarrow \mathrm{Tr}_{\mathcal{H}_{\text{bath}}} \tilde{\rho} =: \Lambda[\rho] \tag{2}$$

which corresponds to the following steps:

1. prepare both the systems that is to be cooled and the bath: the first in a state ρ and the latter in a thermal state ρ_b;
2. let the full system evolve for some time $t > 0$. Here $U(t) = e^{-itH}$ for time-independent Hamiltonians resp. $U(t) = \mathcal{T} \exp\left(-i \int_0^t dt' H_{\text{cooling}}(t')\right)$ for the general case (in natural units where $\hbar = 1$);
3. discard the bath.

The specific protocol of Ding et al. [2026] has several advantages: the bath consists only of one qubit and convergence to the thermal state is obtained by repeated interaction with this bath. The protocol Eq. (2) is a quantum channel with the Gibbs state ρ_β as an approximate fixed point: $\Lambda[\rho_\beta] \approx \rho_\beta$. The channel only relies on forward Hamiltonian resp. effective Lindbladian evolution in the perturbative regime of weak coupling. Although its dynamics are far richer than purely unitary evolution, it is amenable to numeric implementation. Note that the last step in Eq. (2) will in general produce a density matrix rather than a pure state even if the initial state ρ has been pure. Thus, if we think of repeated applications of this scheme, none of the involved states necessarily needs to be pure as $\Lambda^n[\rho_0] \to \rho_{\text{fix}} \approx \rho_\beta$.

We report the concrete implementation of the cooling scheme proposed in Ding et al. [2026] using the Pennylane framework which follows the lines of Eq. (2) and validate it against a known system Hamiltonian, the transverse field Ising model (TFIM) Hamiltonian for small

system sizes $N = 4 - 6$, where exact diagonalization is available and the thermal target states

$$\rho_\beta = \frac{e^{-\beta H_0}}{\mathrm{Tr}\, e^{-\beta H_0}} \tag{3}$$

can be explicitly be calculated. The TFIM models a simple yet nontrivial many-body system with interesting internal structure. The tuning parameter, $\frac{J}{h}$, allows to sweep through different regimes whose limits $J \to 0$ resp. $h \to 0$ are easily interpretable. Moreover, an analytical solution is available for every parameter value, including the critical point $\frac{J}{h} = 1$. This is particularly useful for identifying the source of errors in the cooling channel and avoids possible issues related to an approximate reference solution.

Our numerical experiments include spectral studies of the cooling channels and distances resp. fidelities between the channel fixed points and the exact thermal states. We investigate the dependence on cooling parameters such as coupling strength, modes of energy exchange, filter width and shape, interaction time, frequency cutoff and sampling resolution. We compare both the number of necessary iterations to reach a certain fidelity as well as the effective cooling time (rescaled mixing time) against parameter tuples that arise from a gradient-free optimization procedure. We also recover some properties of the underlying TFIM model, like approximate criticality $\frac{J}{h} \approx 1$ from properties of the channel. Finally we compare the dependencies observed in our experiments to theoretical predictions from Ding et al. [2026] and provide some simple regression models for the mixing time.

This report is structured as follows: First, we introduce the theoretical background of the dissipative cooling protocol and describe it in terms of quantum channels. We then discuss the concrete implementation of the scheme and introduce the TFIM that is used as benchmark system. Afterwards, the numerical results are presented and analyzed, in particular with respect to convergence, fixed points and the dependence on the different parameters. Finally, we summarize the main observations and discuss limitations of the approach. We conclude by outlining possible further directions.

1 Cooling as quantum operation

1.1 System-bath interaction

The two-level system utilized as bath is fully described by its eigenenergies: here they are taken as $\pm\omega/2$, where $\omega > 0$ is a frequency parameter. The fiducial state for the bath subsystem is the two-dimensional Gibbs-state $\rho_\beta \propto e^{\frac{\omega}{2}\beta Z}$.

The basic building blocks for the coupling terms are of the form

$$A_S \otimes \sigma_+ + A_S^\dagger \otimes \sigma_-, \tag{4}$$

each of which describes the physical process of transferring energy from the system to the bath and vice versa. There are a few requirements, both technical and heuristic on how the operator A_S is chosen. Most importantly, A_S itself should be treated as parameter and is chosen from a pre-specified set of operators $\mathcal{A}$. To de-clutter the algorithms, $\mathcal{A}$ is usually chosen to be invariant under Hermitian conjugation and sign inversion. Furthermore $\|A\| \leq 1$, in particular the spectra should be universally bounded. It is essential that $\mathcal{A}$ contains operators that are not diagonal in the energy-eigenbasis but connects states from different energies, making cooling possible in the first place. In the case that H_0 exhibits different symmetry sectors, it is essential that $\mathcal{A}$ mixes those sectors, e.g. if we have a translation-invariant chain of spins, the operators from $\mathcal{A}$ should address the sites individually.

The term in Eq. (4) is multiplied with $\alpha f(t)$, where $\alpha > 0$ is fixed and $f(t)$ a time-dependent coupling strength. We require $\int dt\, |f(t)|^2 = 1$, so $f(t)$ resp. its Fourier-transform $\hat{f}(\omega)$ should be thought of as frequency filter and is usually taken as square-root of a Gaussian $\mathcal{N}(0, \sigma^2)$. $\hat{f}(\omega)$ suppresses transitions from the system to the bath and vice versa that correspond to an allowed detuning of Bohr-frequencies from the bath frequency on a scale set by $1/\sigma$. Thus $H_{\text{int}}(t) = H_{\text{int}}^{(\alpha,A,\sigma)}(t)$.

1.2 Trotterization

For the time evolution a duration $T > 0$ is fixed and the fiducial states are evolved from $-T$ to T. Note, that the negative times are not essential and could easily be avoided by redefining f. However, it is beneficial to preserve the symmetry of f via $f(-t) = f(t)$ over the time interval $[-T, T]$: In this case also the Fourier transform $\hat{f}(\omega)$ is again real (and even) and thus allows for an easy interpretation as frequency filter-function. The full Hamiltonian is now time dependent which makes the computation of the time-propagation more intricate:

$$U(t_0 \to t_1) = \mathcal{T} \exp\left(-i \int_{t_0}^{t_1} d\tau \left(H_0 + \alpha f(\tau)(A_S \otimes \sigma_+ + A_S^\dagger \otimes \sigma_-) + H_{\text{bath}}\right)\right). \tag{5}$$

While an analog simulation would produce the correct time evolution intrinsically, on a digital quantum simulator we have to program the time evolution explicitly and do so with a time-

dependent second order Trotterization for some time Δt

$$U(t \to t+\Delta t) \approx \exp\left(-i\int_{t+\frac{\Delta t}{2}}^{t+\Delta t} d\tau H_{\text{int}}(\tau)\right) e^{-i(H_0+H_{\text{bath}})\Delta t} \exp\left(-i\int_{t}^{t+\frac{\Delta t}{2}} d\tau H_{\text{int}}(\tau)\right) \tag{6}$$

and then approximate the integrals by using the value at the midpoint as constant approximation in order to preserve symmetry which allows for a second order schema.

$$U(t \to t+\Delta t) \approx e^{-iH_{\text{int}}(t+\frac{\Delta t}{2})\frac{\Delta t}{2}} e^{-i(H_0+H_{\text{bath}})\Delta t} e^{-iH_{\text{int}}(t+\frac{\Delta t}{2})\frac{\Delta t}{2}}. \tag{7}$$

In the cooling algorithm itself, the interval $[-T, T]$ is divided into M subintervals of length τ and each subinterval is evolved according to Eq. (7). Note, that freezing H_{int} at the subinterval-midpoint is a convenient numerical choice as it is a second order approximation:

$$\int_{t+\frac{\Delta t}{2}}^{t+\Delta t} d\tau H_{\text{int}}(\tau) = \tfrac{\Delta t}{2} H_{\text{int}}(t+\tfrac{\Delta t}{2}) + A\Delta t^2 + O(\Delta t^3) \tag{8}$$

$$\int_{t}^{t+\frac{\Delta t}{2}} d\tau H_{\text{int}}(\tau) = \tfrac{\Delta t}{2} H_{\text{int}}(t+\tfrac{\Delta t}{2}) - A\Delta t^2 + O(\Delta t^3). \tag{9}$$

It is not hard to show that all operators on the right-hand side of Eqs. (8) and (9) commute. Therefore, to argue why Eq. (7) is actually of second order, it suffices to focus on the term

$$e^{-iA\Delta t^2} e^{-i(H_0+H_{\text{bath}})\Delta t} e^{iA\Delta t^2} = e^{-i(H_0+H_{\text{bath}})\Delta t} + i[e^{-i(H_0+H_{\text{bath}})\Delta t}, A]\Delta t^2 + O(\Delta t^4). \tag{10}$$

Finally, expanding $[e^{-i(H_0+H_{\text{bath}})\Delta t} + O(\Delta^2), A] = [I - i(H_0 + H_{\text{bath}})\Delta t, A] = O(\Delta t)$ we see that the error in Eq. (7) is $O(\Delta t^3)$ for one substep of the time-evolution. Since there are $O(1/\Delta t)$ substeps the global error is $O(\Delta t^2)$.

Using a more sophisticated quadrature of f, or even exact integration, would not necessarily improve the order of the overall approximation, but potentially reduce the constant of the error term. However, we will not pursue this particular direction further.

2 Transverse field Ising model

The benchmark system which is used for validation is given by the Hamiltonian

$$H(J,h) = -J \sum_{<i,j>} \sigma_i^{(z)} \sigma_j^{(z)} - h \sum_{i=1}^{N} \sigma_i^{(x)} \tag{11}$$

and models a 1D chain of spins in z-direction in a homogeneous magnetic field h that is aligned transversely with respect to the orientation of the spin-spin interaction. For simplicity we only consider periodic boundary conditions (i.e. the pair $< 1, N >$ is included in the sum) and an even number of sites.

Due to the next-neighbor interaction there exists a global $\mathbb{Z}_2$-symmetry. Flipping each site commutes with the Hamiltonian: for $P = \sigma_1^{(x)} \cdot \ldots \cdot \sigma_N^{(x)}$ we have $P^2 = I$ and $PH(J,h)P^\dagger = H(J,h)$, while for $Q = \sigma_1^{(z)} \cdot \ldots \cdot \sigma_N^{(z)}$ we have $QH(J,h)Q^\dagger = H(J,-h)$.

The relevant parameter of this system is J/h, the ratio of the energy scale of the spin-spin interaction to the potential energy of the spins in the magnetic field. To keep the energy scales of $H(J,h)$ comparable while sweeping J and h, we usually will impose the normalization

$$\begin{aligned} N \cdot J &\to J/\sqrt{J^2+h^2} \\ N \cdot h &\to h/\sqrt{J^2+h^2} \end{aligned} \tag{12}$$

which leaves the underlying physics unchanged and puts energies per spin, gaps and dynamical timescales essentially on the same scale. For positive J, h with $J \ll h$ a configuration with all spins aligned in $|\uparrow\rangle_x$ is favored (paramagnetic phase) and the spectrum is gaped with $\Delta \approx 2|h|$.

For $J \gg h$ the ground state in the thermodynamic limit $N \to \infty$ is twofold degenerate: for $J > 0$ all spins are either aligned in $|\uparrow\rangle_z$ or in $|\downarrow\rangle_z$ direction (ferromagnetic phase) resp. aligned antiferromagnetically for $J < 0$. The gap to the bulk is $\Delta_{\text{bulk}} \approx 2|J-h|$. In presence of a nonvanishing field h, the degeneracy is lifted and the even parity combination of the above mentioned ferromagnetically aligned states is the ground state, whereas the odd parity combination is the first excited state. The splitting between these two state, i.e. the gap to the first excited state is exponentially small in N. The gap to the bulk remains finite even in the thermodynamic limit as long as the parameter J/h is not in the quantum critical region $J \approx h$. Here the gap closes asymptotically for $N \to \infty$ with $\Delta_{\text{bulk}} \approx \frac{1}{N}$. In our numerical analysis ($N = 4 - 8$) the gap $\Delta_N := E_1 - E_0$ is always finite except in the trivial case of $h = 0$. For a detailed finite-size analysis of this system we refer to Oshikawa [2019].

The spectrum of $H(J,h)$ can be computed exactly by mapping the TFIM to a free fermionic system. First one introduces the Jordan-Wigner transformation which relates spin-strings to

fermionic operators:

$$\hat{c}_j = \left(\prod_{l<j} \sigma_l^{(x)}\right) \frac{\sigma_j^{(z)} - i\sigma_j^{(y)}}{2}, \quad \hat{c}_j^\dagger = \left(\prod_{l<j} \sigma_l^{(x)}\right) \frac{\sigma_j^{(z)} + i\sigma_j^{(y)}}{2}. \tag{13}$$

This transforms Eq. (11) to a quadratic fermionic Hamiltonian which then can be diagonalized via a suitable Bogoliubov transformation in the momentum basis. The final result is a free fermionic Hamiltonian which is diagonal in the number operator basis $\hat{n}_k = \hat{\gamma}_k^\dagger \hat{\gamma}_k$ of the corresponding Bogoliubov quasiparticles:

$$H = \sum_k \varepsilon_k(J,h) \left(\hat{n}_k - \frac{1}{2}\right) \tag{14}$$

with dispersion relation

$$\varepsilon_k(J,h) = 2\sqrt{(h - J\cos k)^2 + J^2 \sin^2 k} \quad \text{for} \quad k = \pm\frac{\pi}{N}, \pm\frac{2\pi}{N}, \ldots \tag{15}$$

In particular for $J = h$ one gets

$$H = \sum_k 4J|\sin\tfrac{k}{2}| \left(\hat{n}_k - \tfrac{1}{2}\right). \tag{16}$$

The equivalence of TFIM to a free fermionic Hamiltonian allows for a priori estimates of the integer mixing time of the cooling algorithm. This would be much harder to obtain for other models, see Ding et al. [2026, Appendix F.2]. Therein an $\mathcal{O}(N^2 \log \frac{N}{\varepsilon})$ dependence on system size and tolerated error is asserted. However this approach hinges on the choice $\mathcal{A} = \{\pm\gamma_k, \pm\gamma_k^\dagger\}$ of coupling operators, which are *not* simply Pauli operators on individual sites but rather nonlocal combinations thereof. In our implementation we aim to have both operations P and Q available and thus are more interested in sets of coupling operators that contain $\sigma_1^{(x)}, \ldots, \sigma_N^{(x)}$ *and* $\sigma_1^{(z)}, \ldots, \sigma_N^{(z)}$.

3 Parameter dependence

Let us quickly review the parameters we encountered so far:

coupling strength α determines the magnitude of the interaction part in the cooling Hamiltonian, i.e. the maximal rate at which cooling processes mediated by the bath can take place. To allow for the description of the cooling process via perturbative Lindblad dynamics, α is treated as small parameter in Ding et al. [2026]. However, larger coupling allows for higher energy transfer from the system to the bath and therefore increases cooling. In

Wang and Ding [2025] the validity of the cooling scheme is established also for $\alpha = O(1)$. The numerical experiments of this work underscore these findings.

temporal width σ denotes the standard deviation of a Gaussian filter function in time. σ tunes the scale at which transitions from bath to system and vice versa with larger than $\frac{1}{\sigma}$ detuning are suppressed. A safe choice is $\hbar/\sigma \lesssim \|H_0\|$ resp. the energy bandwidth of the system that is to be cooled.

frequency cutoff $\omega_{\max}$ determines the range of transition frequencies that are sampled. In Ding et al. [2026] this cutoff is necessary since the transition frequencies are sampled uniformly

$$g(\omega) = \frac{1}{\omega_{\max}} \mathbb{1}_{[0,\omega_{\max}]}(\omega). \tag{17}$$

The cutoff has to be large enough to include the relevant Bohr frequencies of the system. A safe choice is the energy bandwidth $\omega_{\max} \geq 2\|H\| \geq E_{\max} - E_{\min}$. In practice, however, it is not necessary to include *all* but sufficiently many frequencies. Therefore a cutoff at the median transition frequency serves as a good starting point. It should be noted that a uniform distribution is merely convenient, but not necessarily the optimal choice. In light of Fig. 1, $\sigma = 4$ is certainly safe for TFIM $N = 4 - 6$, even when accounting for discretization artefacts.

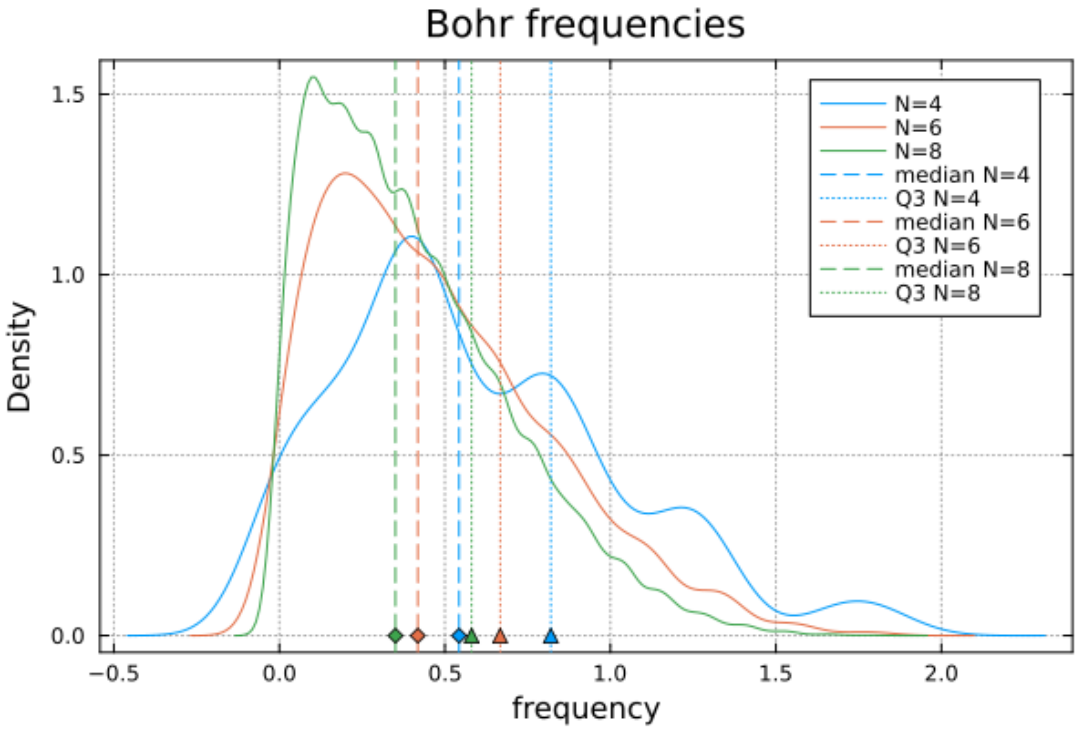


Figure 1: TFIM for $h = 1.2$, $J = 1$ and normalization factor $N\sqrt{J^2 + h^2}$.

interaction time T determines the finite time interval $[-T, T]$ during which system and bath interact. During the interaction, the coupling constant is modulated with the Gaussian f. Increasing T decreases the error due to ignoring f outside $[-T, T]$ but increases the simulation time necessary to evolve the combined Hamiltonian H_{cooling} which itself is subject to discretization errors. In Ding et al. [2026] it is assumed that $T \gg \sigma$ in order to guarantee convergence.

Trotter step size τ and order determine the discretization scheme of the time evolution on the interval $[-T, T]$. There are $M = \left\lceil \frac{2T}{\tau} \right\rceil$ (meaning the smallest integer larger or equal than $\frac{2T}{\tau}$) Trotter steps necessary. The scheme chosen in Eq. (7) is of second order and thus avoids negative weights in the approximation. Decreasing τ reduces the Trotter error but increases the circuit depth and number of numerical evaluations. Consequently, τ has to resolve both the system dynamics generated by H_0 and the system-bath interaction. As heuristic τ should be small compared to the times scale set by the free evolution of the system Hamiltonian and the interaction with the bath.

frequency spacing $\Delta\omega$ influences error of the quadrature rule for the integration of the bath frequency sampling density $g(\omega)$. In the numerical implementation of the algorithm $[0, \omega_{\max}]$ is replaced by a discrete frequency grid. The spacing has to be sufficiently small to resolve the relevant Bohr frequencies. On the other hand, a too coarse frequency grid can miss relevant transitions and therefore change both the cooling rate and the averaged channel.

coupling operators $\mathcal{A}$ determine which transitions between energy eigenstates can be induced by interactions with the bath. In each application of the cooling channel, an operator A_S is sampled from $\mathcal{A}$. The set has to connect relevant parts of the spectrum. Otherwise conserved quantities or disconnected invariant subspaces can prevent convergence to a unique thermal state. For spin systems, a natural choice is a set of local Pauli operators, provided not all of them commute with the system Hamiltonian: We implemented $\mathcal{A}_{XZ} = \{X_i, Z_i\}$ as standard set as well as $\mathcal{A}_{XYZ} = \{X_i, Y_i, Z_i\}$ and $\mathcal{A}_d = \{X_i, Z_i Z_j\}$. The latter set being of particular interest with respect to the duality of TFIM, however the numerical experiments in this report are restricted to $\mathcal{A} = \mathcal{A}_{XZ}$.

4 Lindbladian approximation

Ding et al. [2026] show that the action of the cooling channel $\mathcal{E}$ can be approximated by a unitary time evolution of the system plus a Lindbladian action:

$$\mathcal{E}[\rho] \approx \mathcal{E}_{\mathcal{M}}[\rho] = \mathcal{U}(T) \circ \exp\left(\mathcal{M}\alpha^2\right) \circ \mathcal{U}(T), \tag{18}$$

where

$$\mathcal{U}(T)[\rho] = \exp(-iH_0T)\rho\exp(iH_0T) \tag{19}$$

and $\mathcal{M}$ is the generator of a carefully constructed Lindbladian that preserves the Gibbs state ρ_β. It is shown that this approximation is arbitrarily close to an Lindbladian channel that has error $O(\alpha^4T^4/\sigma^2)$. In particular, we see that the Lindbladian part of the evolution evolves with an effective time increment of α^2. Since

$$\mathcal{U}(T)[\rho_\beta] = \frac{1}{\mathcal{Z}}\exp(-iH_0T)\exp(-\beta H_0)\exp(iH_0T) = \frac{1}{\mathcal{Z}}\exp(-\beta H_0) = \rho_\beta \tag{20}$$

leaves Gibbs states invariant and the Lindbladian generator fulfills $\mathcal{M}[\rho_\beta] \approx 0$ by construction, the right-hand side of Eq. (18) has ρ_β as an approximate fixed point. Hence, as long as the approximation Eq. (18) is valid, ρ_β is an approximate fixed point of the cooling channel $\mathcal{E}[\cdot]$. In Ding et al. [2026] it is proven that under certain conditions a matrix norm can be chosen such that $\mathcal{E}_\mathcal{M}$ is a contraction with respect to this norm, hence establishing uniqueness of the fixed point. However, uniqueness of the exact channel $\mathcal{E}$ does not necessarily follow.

4.1 Dyson expansion and dissipators

To construct the Lindbladian we first resolve the system-bath interaction in Bohr-frequency components with respect to the H_0 eigenbasis

$$A = \sum_\nu \overbrace{\sum_{E_i - E_j = \nu} \langle i|A|j\rangle |i\rangle\langle j|}^{A_\nu}. \tag{21}$$

We see, that in the interaction picture these operators simply evolve as

$$A_I(t) = e^{itH_0}Ae^{-itH_0} = \sum_\nu e^{i\nu t}A_\nu. \tag{22}$$

The interaction Hamilitonian can be written as

$$H_I = \alpha f(t)\sum_\nu e^{i\nu t}A_\nu \otimes e^{i\omega t}\sigma_+ + \text{h.c.} \tag{23}$$

Therefore the Dyson expansion of the time propagator from $-T$ to T in powers of α is

$$\begin{aligned} U_{\text{int}}(T) &= \mathcal{T}\exp(-i\int_{-T}^{T} dt' H_I(t')) \\ &= I - \underbrace{i\int_{-T}^{T} dt' H_I(t')}_{=-\alpha A_1} - \underbrace{\int_{-T}^{T}\int_{-T}^{t'} dt'dt'' H_I(t')H_I(t'')}_{-\alpha^2 A_2} + o(\alpha^2). \end{aligned} \tag{24}$$

The action on density matrices is

$$\rho \mapsto U_{\text{int}}(T)\rho U_{\text{int}}^\dagger(T) = \rho + \alpha\left(A_1\rho + \rho A_1^\dagger\right) + \alpha^2\left(A_1\rho A_1^\dagger + A_2\rho + \rho A_2^\dagger\right) + o(\alpha^2). \tag{25}$$

For $\rho = \rho_S \otimes \rho_\beta$ it turns out, that the first order term vanishes and only the second order contributions survive tracing out the bath. The second order terms combine to dissipator contributions in the Lindbladian:

$$\begin{aligned} &(A(\nu)\otimes\sigma_+)\,\rho\left(A^\dagger(\mu)\otimes\sigma_-\right)\int_{-T}^{T}\int_{-T}^{T} dtdt' e^{i(\nu+\omega)t}e^{-i(\mu+\omega)t'}f(t)f(t') && \text{resp. h.c.} \\ -&\rho\left(A^\dagger(\nu)A(\mu)\otimes\sigma_-\sigma_+\right)\int_{-T}^{T}\int_{-T}^{t} dtdt' e^{-i(\nu+\omega)t}e^{i(\mu+\omega)t'}f(t)f(t') && \text{resp. h.c.} \\ -&\left(A^\dagger(\nu)A(\mu)\otimes\sigma_-\sigma_+\right)\rho\int_{-T}^{T}\int_{-T}^{t} dtdt' e^{-i(\nu+\omega)t}e^{i(\mu+\omega)t'}f(t)f(t') && \text{resp. h.c.} \end{aligned} \tag{26}$$

After tracing out the bath state ρ_β each contribution gets weighted either by p_0, the Gibbs weight of $|0\rangle\langle 0|$, for $\sigma_-\sigma_+$ and $\sigma_+\rho\sigma_-$ processes or by p_1, the Gibbs weight of $|1\rangle\langle 1|$, for the hermitian conjugate $\sigma_+\sigma_-$ and $\sigma_-\rho\sigma_+$process. So, the Bohr components of $A \in \mathcal{A}$ get weighted according to the frequency filter f and then contribute as jump operators to the second order Lindbladian approximation:

$$V_A(\omega) = \sum_\nu \int_{-T}^{T} dt f(t) e^{i(\nu+\omega)t} A_\nu \tag{27}$$

with

$$\mathcal{D}_A[\rho_S] = \langle 0|\,\rho_\beta\,|0\rangle\left(V_A\rho_S V_A^\dagger - \frac{1}{2}\{\rho_S, V_A^\dagger V_A\}\right) \tag{28}$$

$$\mathcal{D}_{A^\dagger}[\rho_S] = \langle 1|\,\rho_\beta\,|1\rangle\left(V_A^\dagger\rho_S V_A - \frac{1}{2}\{\rho_S, V_A V_A^\dagger\}\right) \tag{29}$$

Note that A always corresponds to the σ_+ factor and $A^\dagger$ to the σ_- factor in Eq. (4). Also, along with the dissipators Eqs. (28) and (29) also imaginary contributions, the so-called Lamb-shift terms, arise. Since we are mainly interested in the dissipative processes of the quantum channel we don't follow this direction and instead refer to Ding et al. [2026, Appendix C].

4.2 Transition rates

In particular, for $T \gg \sigma$ the matrix elements in the energy basis of H_0 can be approximated by extending the integration limits:

$$\begin{aligned}\langle i|\, V(\omega)\, |j\rangle &= \sum_{\nu} \int_{-T}^{T} dt f(t) e^{i(\nu+\omega)t} \langle i|\, A\, |j\rangle\, \delta_{E_i - E_j, \nu} = \langle i|\, A\, |j\rangle \cdot \int_{-T}^{T} dt f(t) e^{i(E_i - E_j + \omega)t} \\ &\approx \langle i|\, A\, |j\rangle \cdot \sqrt[4]{8\pi\sigma^2} \int_{-\infty}^{\infty} dt \frac{1}{\sqrt{2\pi \cdot 2\sigma^2}} e^{-\frac{t^2}{4\sigma^2}} e^{i(E_i - E_j + \omega)t} \\ &= \sqrt[4]{8\pi\sigma^2} e^{-\sigma^2 (E_i - E_j + \omega)^2} \langle i|\, A\, |j\rangle\,. \end{aligned} \tag{30}$$

A careful inspection of the dissipator terms of Eqs. (28) and (29) reveals that the dissipator contribution of $A \in \mathcal{A}$ at frequency ω is responsible for the transfer of H_0 eigenstates with transition strength

$$T_{i \leftarrow j}(\omega) = |\, \langle i|\, V(\omega)\, |j\rangle\, |^2 \approx \sqrt{8\pi}\sigma |\, \langle i|\, A\, |j\rangle\, |^2 e^{-2\sigma^2 (E_i - E_j + \omega)^2} \qquad (i \neq j). \tag{31}$$

Consequently, the population dynamics of the Lindbladian dynamics in the H_0 eigenbasis play out with transition rates

$$\begin{aligned} t_{i \leftarrow j} \approx \sqrt{8\pi}\sigma \,(\, &\mathbb{E}_{\mathcal{A}} |\, \langle i|\, A\, |j\rangle\, |^2 \cdot \mathbb{E}_{[0,\omega_{\max}]} p_0(\omega) e^{-2\sigma^2 (E_i - E_j + \omega)^2} + \\ &\mathbb{E}_{\mathcal{A}} |\, \langle i|\, A^\dagger\, |j\rangle\, |^2 \cdot \mathbb{E}_{[0,\omega_{\max}]} p_1(\omega) e^{-2\sigma^2 (E_i - E_j - \omega)^2}\,) \end{aligned} \qquad (i \neq j) \tag{32}$$

and the rates $t_{i \leftarrow i} = -\sum_{j \neq i} t_{j \leftarrow i}$ are computed asserting balance. Thus, if the matrix elements $\langle i|\, A\, |j\rangle$ are available, e.g. by exact diagonalization for small N, this may serve as numerical benchmark on how close the Lindbladian approximation is to the actual cooling channel:

$$p_{i \leftarrow j} = \langle i|\, \mathcal{E}\, [|j\rangle\, \langle j|]\, |i\rangle = \delta_{ij} + \alpha^2 t_{i \leftarrow j} + o(\alpha^2). \tag{33}$$

Thus expanding Eq. (32) in powers of σ implies that

$$\frac{p_{i \leftarrow j}}{t_{i \leftarrow j}} = c(\omega_{\max}, \mathcal{A}) \cdot \alpha^2 + o(\alpha^2) \qquad (i \neq j). \tag{34}$$

5 Implementation as quantum circuit

In order to implement concrete realizations of the quantum channel, we used the PENNYLANE python package. Here 'concrete' means, that the total cooling Hamiltonian H is built using a specific jump operator $A \in \mathcal{A}$ and a specific Bohr frequency $\omega \in [0, \omega_0]$ and evolved from time $t = -T$ to time $t = T$. The Trotterization from Eq. (7) is directly implemented into the circuit,

meaning that once the total evolution time $2T$ and the sub-step time τ are chosen, the quantum circuit is built according to these specifications. It consists of $M+1$ layers, each effecting a time increment of τ. In Fig. 2 a circuit which implements two cooling steps is shown, each of which uses individual Trotter steps of length $\tau = T/2$. Here $W(\tau)$ denotes the time-evolution of H_{int}, whereas the time evolution of H_{bath} is implemented directly as Z-rotation, and the time evolution of H_0 is obtained by Trotterization with the same step size τ.

After each cooling step, the auxiliary qubit is measured and prepared in the same Gibbs state of the two-level system. The PENNYLANE framework offers two approaches on how to implement this reset:

- we choose the initial state of the auxiliary qubit probabilistically according to the statistics of the Gibbs state, i.e.

$$p_0(\omega) = \langle 0| \rho_\beta |0\rangle = \frac{1}{1+e^{-\beta\omega}} \quad \text{resp.} \quad p_1(\omega) = \langle 1| \rho_\beta |1\rangle = \frac{1}{1+e^{\beta\omega}}. \tag{35}$$

- we run the simulation in mixed mode, meaning states are not assumed to be pure but mixed instead. In this case we can implement the constant 2-dimensional quantum channel

$$\rho \mapsto \sum_{i,j} \underbrace{\sqrt{p_i}\,|i\rangle\langle j|}_{K_{ij}} \rho \underbrace{\sqrt{p_i}\,|j\rangle\langle i|}_{K_{ij}^\dagger} = \sum_i p_i\,|i\rangle\langle i|\ \mathrm{Tr}\rho = \rho_{\left(\beta\frac{\omega}{2}\right)}, \tag{36}$$

  where we have used the channel representation with Kraus-operators. This can be implemented directly in PENNYLANE.

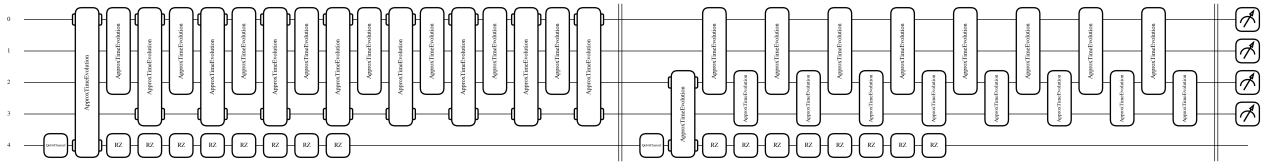


Figure 2: Quantum circuit constructed using the PENNYLANE framework.

We note that the simulation of the quantum circuit makes it necessary to decide in advance how many iteration steps shall be undertaken. In a physical quantum computer the gate sequence which constitutes one cooling step would instead be applied iteratively (e.g. a fixed sequence of laser pulses) in a trapped ion or neutral atom setup.

Furthermore, for the simulation all iteration steps have to be pre-determined, meaning a specific sequence $(A_n, \omega_n) \in \mathcal{A} \times [0, \omega_{\max}]$ has to be chosen in advance. In the present implementation this is done with a random seed, in order to allow for reproducibility.

6 Simulation using different channel representation

A quantum channel is a map that maps density matrices to density matrices, i.e. $\mathcal{E} : \mathbb{C}^{2^N \times 2^N} \to \mathbb{C}^{2^N \times 2^N}$. If we view $\mathcal{E}$ as a transformation of matrices, it is common to use the name superoperator instead. $\mathcal{E}$ can then be treated as a $4^N \times 4^N$ matrix. The identification $\mathbb{C}^{2^N \times 2^N} \cong \mathbb{C}^{4^N}$ is called „vectorization", cf. Watrous [2018].

6.1 Kraus representation

Every quantum channel has also - at least one - Kraus representation, i.e. can be written as

$$\mathcal{E}[\rho] = \sum_i K_i \rho K_i^\dagger. \tag{37}$$

Since the system's state space is $\mathcal{H} = \mathcal{H}_0 \otimes \mathcal{H}_b$, where the bath's state space is just two-dimensional, this allows for some simplification: Let us denote by $U = U(t_m, t_{m+1})$ the time evolution of the composite system including the bath for a particular time interval. We can write

$$U = \begin{pmatrix} U_{00} & U_{01} \\ U_{10} & U_{11} \end{pmatrix} = \sum_{i,j} U_{ij} \otimes |i\rangle_b \langle j|_b \,, \tag{38}$$

where $U_{ij} = \langle i|_b U |j\rangle_b$ is an operator $\mathcal{H}_0 \to \mathcal{H}_0$, i.e. a $2^N \times 2^N$ matrix. Applying $U\rho U^\dagger$ to a density matrix of the form $\rho = \rho_0 \otimes \sum_k p_k |k\rangle_b \langle k|_b$, we get

$$\begin{aligned} \mathcal{E}[\rho] = \mathrm{Tr}_{\mathcal{H}_{\mathrm{bath}}} U\rho U^\dagger &= \mathrm{Tr}_{\mathcal{H}_{\mathrm{bath}}} \sum_{i,j,i',j',k} p_k U_{ij} \rho_0 U^\dagger_{i'j'} \otimes |i\rangle_b \langle j|_b |k\rangle_b \langle k|_b |j'\rangle_b \langle i'|_b & (39) \\ &= \mathrm{Tr}_{\mathcal{H}_{\mathrm{bath}}} \sum_{i,i',k} p_k U_{ik} \rho_0 U^\dagger_{i'k} \otimes |i\rangle_b \langle i'|_b & (40) \\ &= \sum_{i,k} p_k U_{ik} \rho_0 U^\dagger_{ik} = \sum_{i,k} \sqrt{p_k} U_{ik} \rho_0 (\sqrt{p_k} U_{ik})^\dagger, & (41) \end{aligned}$$

i.e. four Kraus operators. This involves only matrix products of $2^N \times 2^N$ matrices, which is advantageous with respect to multiplying $2^{N+1} \times 2^{N+1}$ matrices.

6.2 Superoperator representation

We implement a channel $\mathcal{E}$ by so called row-wise vectorization convention: for a $d \times d$ matrix X, with $d = 2^N$, its rows are stacked into a column vector,

$$\mathrm{vec_r(X)} = (\mathrm{X_{11}}, \ldots, \mathrm{X_{1d}}, \mathrm{X_{21}}, \ldots, \mathrm{X_{dd}})^{\mathrm{T}}. \tag{42}$$

In this convention,

$$\mathrm{vec}_r(AXB^\dagger) = (B^* \otimes A)\mathrm{vec}_r(X), \tag{43}$$

where * denotes entrywise complex conjugation. A quantum channel is therefore represented by a $d^2 \times d^2$ matrix S satisfying

$$\mathrm{vec}_r(\mathcal{E}[\rho]) = S_\mathcal{E}\mathrm{vec}_r(\rho). \tag{44}$$

In particular, for a Kraus representation $\mathcal{E}[\rho] = \sum_i K_i \rho K_i^\dagger$, this gives

$$S_\mathcal{E} = \sum_i K_i \otimes K_i{}^*. \tag{45}$$

6.3 Averaged channel

The cooling algorithm proposed in Ding et al. [2026] is probabilistic in the sense that end-to-end guarantees apply to the *average* channel, where an expectation value is taken both with respect to the jump operators and the Bohr frequencies:

$$\mathcal{E}\left[\rho\right] := \mathbb{E}_{A,\omega}\Lambda(A,\omega)\left[\rho\right]. \tag{46}$$

In choosing the underlying probability distributions on $\mathcal{A}$ resp. $[0, \omega_{\max}]$, we follow Ding et al. [2026] and choose uniform distributions on both sets. For $\mathcal{A}$ this seems to be the canonical choice - at least if the jump operators correspond to physical sites and the Hamiltonian is translation invariant.

In this work we focus on the set $\mathcal{A} = \{\pm\sigma_i^{(x)}, \pm\sigma_i^{(z)} : i = 1, \dots N\}$ and the uniform probability assigning each operator $1/(4N)$. As simple choice for the set of Bohr frequencies $[0, \omega_{\max}]$ we choose a uniform distribution as well. Here, however, there is a priori no need to exclude other distributions and it could turn out that e.g. distributions that are skewed towards higher frequencies (not considered here) are advantageous.

In order to realize the averaged channel $\mathcal{E}[\cdot]$ numerically, we approximate using numerical integration

$$\frac{1}{\omega_{\max}}\int_0^{\omega_{\max}} d\omega\, \Lambda(A,\omega) = \frac{1}{k}\sum_{i=1}^{k}\Lambda(A,\omega_i) + O(k^{-2}) \tag{47}$$

by some quadrature formula, e.g. by the midpoint-formula where $\omega_i = \frac{2i-1}{2k}\omega_{\max}$. The average

channel is thus approximated by

$$\mathcal{E} \approx \frac{1}{4Nk} \sum_{A \in \mathcal{A}} \sum_{i=1}^{k} \Lambda(A, \tfrac{2i-1}{2k}\omega_{\max}) \tag{48}$$

up to order $O(k^{-2})$. Thus the number of individual contributions for a Hamiltonian on N sites scales with $4N\,k$. To keep the computational demand affordable, we choose $k = 10$. For computing the average channel for $N = 4, 6$ we then have to run trajectories of length 160 resp. 240 each.

7 Numerical Experiments

In this section, we numerically investigate the cooling channel and its dependence on the relevant model and channel parameters. We first consider a range of transverse-field ratios h/J, examining the convergence rate, the accuracy of the fixed point, and the spectrum for different coupling strengths. This sweep also includes a comparison between the dynamics of the exact channel and the corresponding Lindbladian approximation. We then study the dependence on the remaining channel parameters, σ and β and use these results to discuss the main numerical trade-offs. We mostly use $N = 4$ spins, only for the discussion of the system size dependence we additionally rely on $N = 6$ results.

7.1 Model and channel parameters

As first numerical experiment we study how the cooling channel behaves as the transverse field of the TFIM is varied. We sweep the ratio h/J from the ferromagnetic regime $h/J < 1$, through the critical region around $h/J = 1$, into the paramagnetic regime $h/J > 1$. To avoid changing the overall energy scale of the system simultaneously, J and h are normalized according to Eq. (12). The normalization here serves also a numerical purpose: would we not normalize, either the energy scale for larger h/J would meaningful exceed $[0, \omega_{\max}]$ or $\omega_{\max}$ would have to be adjusted to accommodate also those large energies, resulting in an artificially coarse frequency spacing $\Delta\omega$, that would distort results for smaller h/J. For each ratio h/J of the sweep the averaged cooling channel $\mathcal{E}$ is constructed explicitly according to Eq. (46). Its fixed point ρ^* is obtained from the eigenvector corresponding to the eigenvalue $\lambda = 1$. We compare five different coupling strengths $\alpha = 0.25$, 0.5, 0.75, 1.0, and 1.5, while all remaining parameters of the cooling channel are kept fixed: $T = 25$ with $\tau = 0.1$ (i.e. 500 sub-steps per Trotterization), $\sigma = 2$ ensuring that the temporal window $[-T, T]$ is large compared to σ. The frequency cutoff is chosen well above the systems bandwidth at $\omega_{\max} = 4$ and the inverse temperature is set to $\beta = 1$.

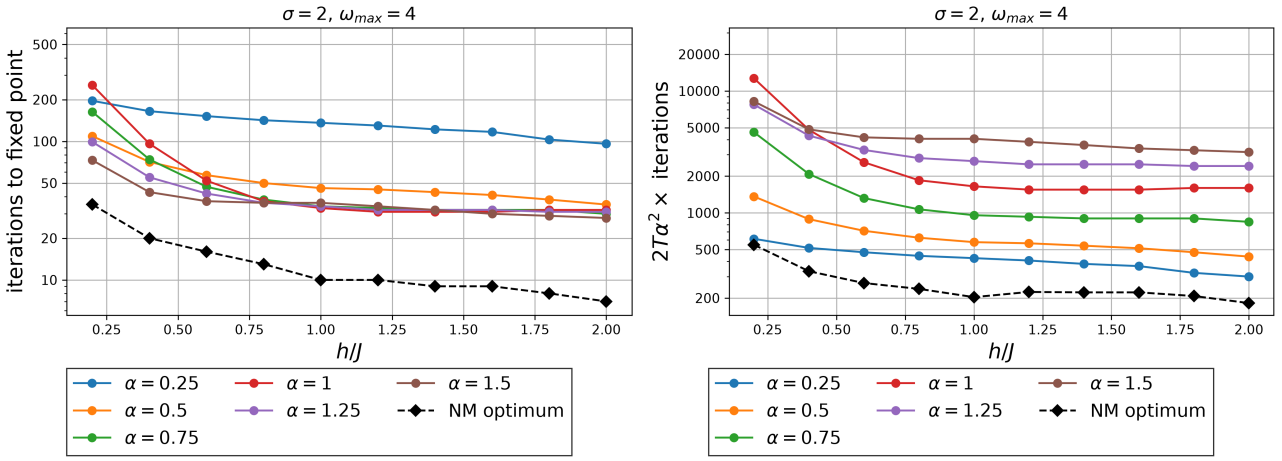
Starting from the pure state $\rho_0 = |0000\rangle\langle 0000|$, we iterate until the trace-distance

$$D(\mathcal{E}^n[\rho_0], \rho_*) = \tfrac{1}{2} \|\mathcal{E}^n[\rho_0] - \rho_*\|_1 \tag{49}$$

is less than $\varepsilon = 0.05$, which defines our convergence criterion. We note that this notion differs from the one used in Ding et al. [2026] insofar as no supremum over initial states is taken. Instead, as is common in quantum computation, the initial state $|0000\rangle\langle 0000|$ is fixed.

7.2 Accuracy and convergence

When given a specific Hamiltonian, a desired temperature, and an ϵ target accuracy, the main question regarding a cooling protocol is obviously: *how long will it take?* We can measure either the integer number of iterations N, the corresponding evolution time of the unitary dynamics $2TN$ or the corresponding evolution time of the Lindbladian dynamics $2T\alpha^2 N$.



(a) Number of iterations.

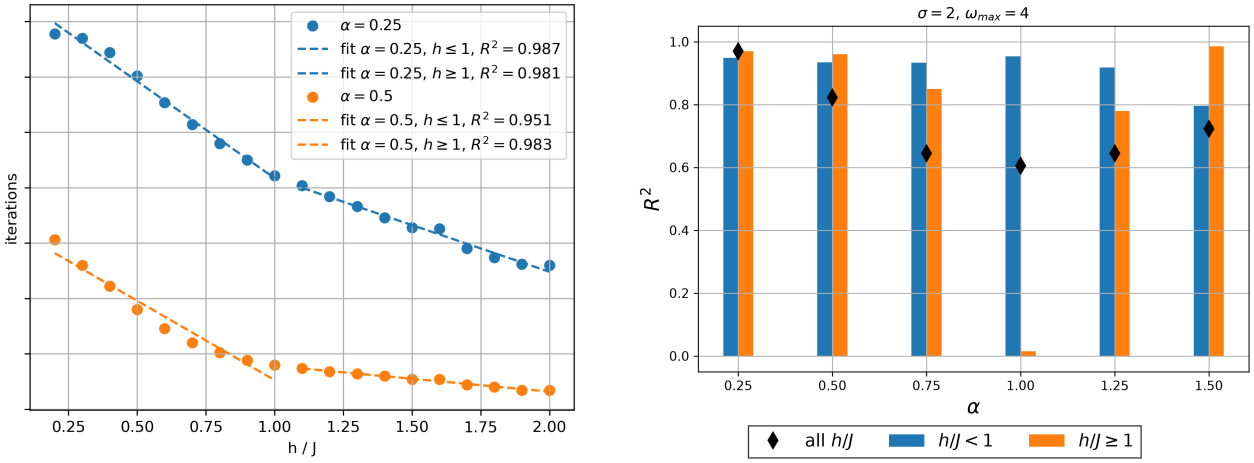
(b) Effective Lindbladian evolution time $2T\alpha^2 \cdot N$.

Figure 3: Convergence within trace-distance $\varepsilon = 0.05$ to channel fixed-point for the TFIM with $N = 4$ cooling channels, for various values of the coupling-strength α. NM refers to parameter configurations obtained using the gradient free Nelder-Mead optimizer.

In Fig. 3 we see that the number of iterations necessary to get ε close to the fixed point decreases almost monotonically with h/J, which implies that convergence is reached faster in the paramagnetic regime than in the ferromagnetic regime. In particular, there seems to be nothing special about the critical region $h/J \approx 1$. (Note that the Hamiltonians are all normalized according to Eq. (12) in order to avoid different energy scales of the systems.) At the same time we observe a quite different qualitative behavior for the five tested coupling strengths $\alpha = 0.25 - 1.5$. While in general the number of necessary iterations decreases for larger α, the functional dependence changes significantly. The smallest $\alpha = 0.25$ exhibits an almost exponential decay with respect to h/J (linear in the semi-logarithmic plot Fig. 3a) whereas

$\alpha \geq 0.75$ display a clearly nonlinear dependence approaching a constant baseline. $\alpha = 0.5$ is qualitatively closer to the linear case, but seems to have different slopes in the para- resp. ferromagnetic region, see Fig. 4.

The number of iterations required to reach the fixed point does not decrease uniformly across the parameter space. In the nonferromagnetic regime, higher coupling imply faster convergence, although we observe only a marginal improvement from $\alpha = 0.75$ onwards. Qualitatively, a transition from log-linear to nonlinear takes place for increasing coupling strength α.



(a) Piecewise linear fits in the semilog plot for $h/J \leq 1$ and $h/J \geq 1$.

(b) Fit quality (R^2) of the semilog regressions in the $h/J \leq 1$ and $h/J \geq 1$ regions.

Figure 4: Exponential fits to the number of iterations required for convergence to the fixed point of the TFIM cooling channel ($N = 4$), for a h/J sweep over 10 points from 0.2 to 2.0 and varying coupling strengths α.

It is important to keep in mind that the fixed point of a given channel is not necessarily the desired Gibbs state, thus it is necessary to quantify this distance first. Fig. 5 validates the convergence of the implemented cooling against the exact thermal state. Here we clearly see a tradeoff between convergence speed and accuracy: convergents for small α channels stay closer to the desired target state than convergents for larger α. In the concrete numeric experiment only $\alpha = 0.25$ and $\alpha = 0.5$ guarantee a 2ε-closeness of the cooled state to the Gibbs state. We also see a different qualitative behavior for small and large values of α with respect to the parameter h/J: small α need fewer iterations the farther we are in the paramagnetic region, while large α require more iterations.

Using the Nelder-Mead method, we performed a gradient-free minimization in order to determine the number of necessary iterations across the parameter space, with a threshold of $\varepsilon = 0.05$ to the Gibbs state. This optimization handily surpasses every sweep parameter choice, where the improvement is more pronounced in the paramagnetic regime than in the ferromagnetic one. Interestingly, it turns out that the optimal coupling strength α is not at weak coupling

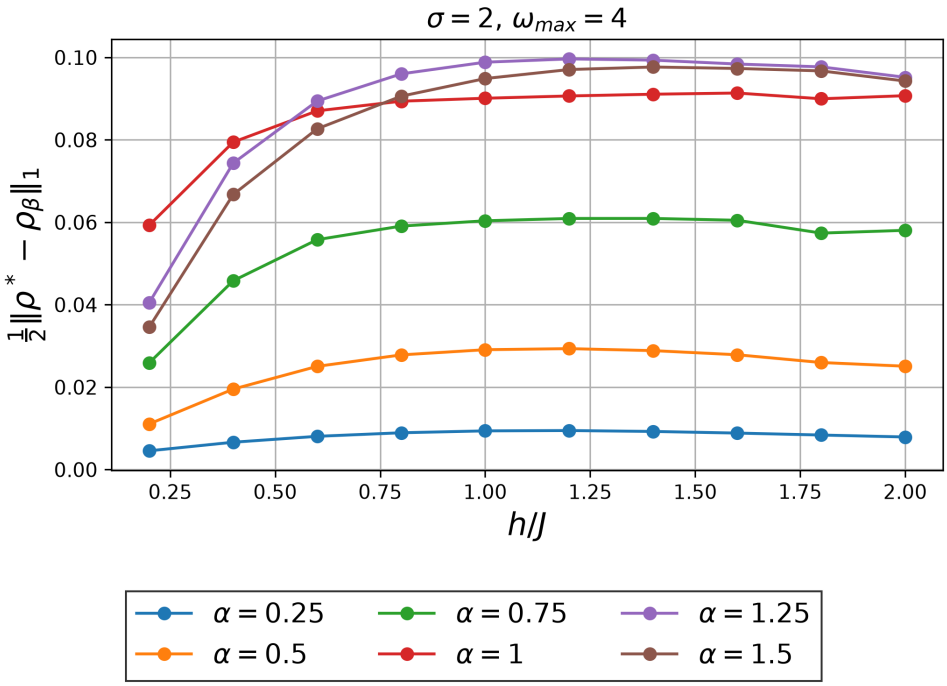


Figure 5: Trace distance between the channel fixed point and the Gibbs state for the TFIM cooling channel with $N = 4$, $\beta = 1$, $\sigma = 2$, and $\omega_{\max} = 4$, for different coupling strengths.

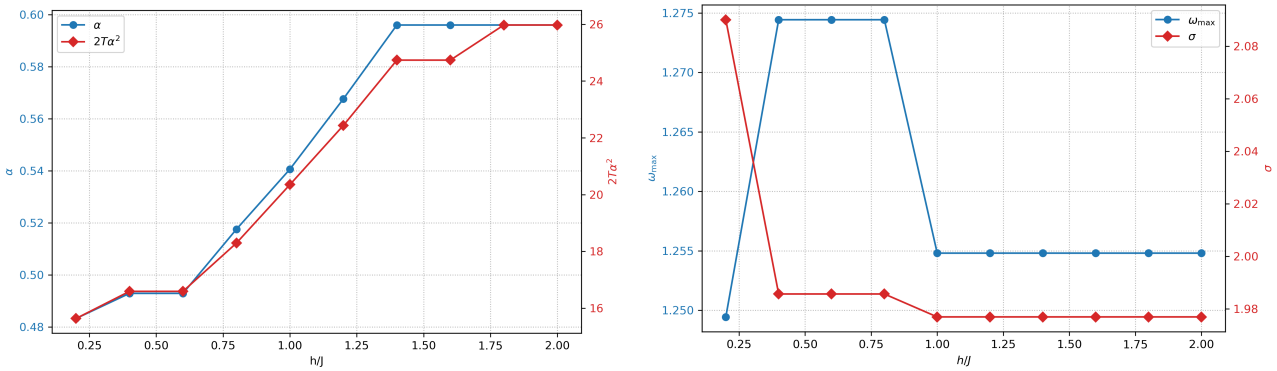
but around $0.5 - 0.6$, see Fig. 6a. The $\omega_{\max}$ cutoff values that lead to minimal iteration count are surprisingly small at only roughly half the bandwidth of the system. The temporal width is almost constant $\sigma \approx 2$. The small variance of these parameters across h/J may be taken as indication that the iterations count is barely affected by their choice and that the specific values originate more from the warm start of the algorithm (taking the optimal parameters of the previous run as starting parameters of the next run).

In conclusion: the h-sweep eludes to the main trade-off in the numerics very clearly: the weak-coupling regime results in the most accurate approximation of the Gibbs state, but at the cost of many channel iterations. At intermediate coupling strengths, one can lower the iteration count significantly while still keeping the fixed-point error relatively small.

7.3 Channel spectrum

In order to study the spectral properties of the channel with respect to its fixed-point behavior we introduce the quantity

$$\Delta_{\text{sep}} = \min_{\lambda \neq 1} |1 - \lambda|. \tag{50}$$



(a) Number of iterations N and effective Lindbladian evolution time $2T\alpha^2 \cdot N$.

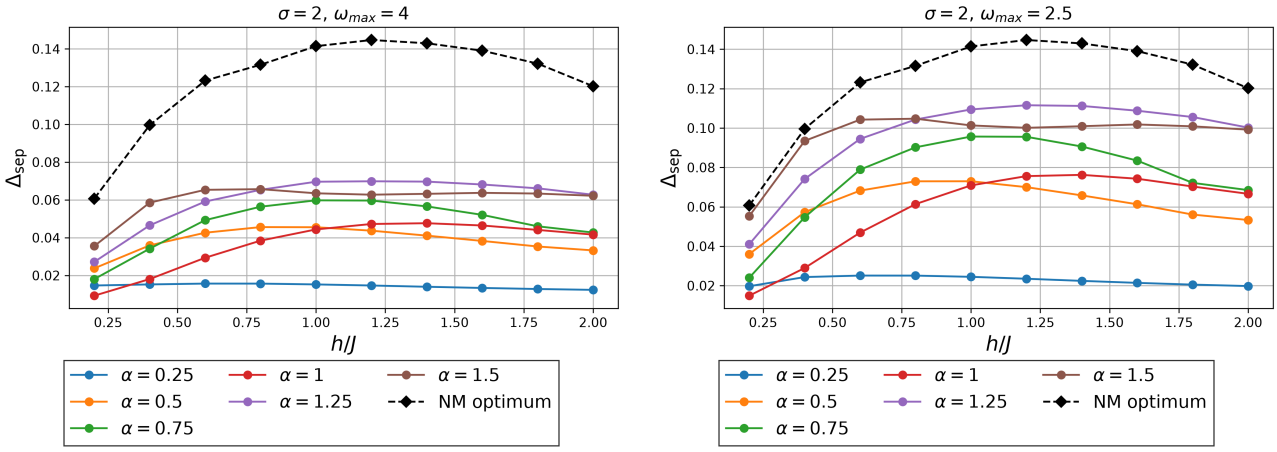
(b) Frequency cutoff $\omega_{\max}$ and temporal width σ.

Figure 6: Parameters for Nelder-Mead optimized channels with up to 100 iterations for each h/J point and until $\varepsilon = 0.05$ closeness to ρ_β is established. The sweep was performed from smaller to larger h/J and parameters of the optimizer of the previous point were used as initial parameters for the next point.

Note that Eq. (50) is in general different from the spectral *gap*

$$\Delta_{\text{gap}} = 1 - \max_{\lambda \neq 1} |\lambda| \leq \Delta_{\text{sep}}. \tag{51}$$

The two notions coincide only for real nonnegative subleading eigenvalues. By perturbative arguments we know that - at least in the weak-coupling regime - the channel is guaranteed to have an almost thermal fixed point, cf. Ding et al. [2026, Theorem 9]. For this reason we are mostly interested in the local behavior around this fixed point rather than in the global gap.



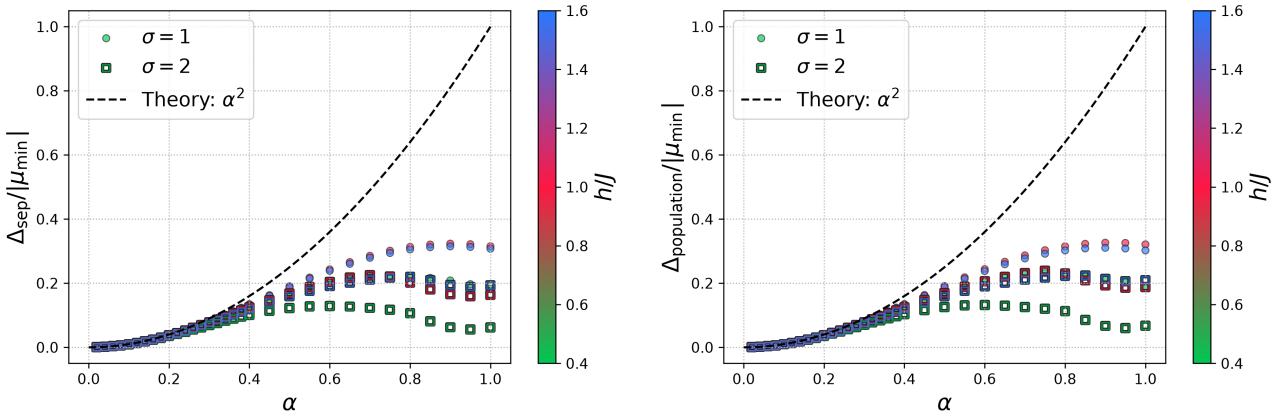
(a) Higher frequency cutoff $\omega_{\max} = 4$ resulting in $\Delta\omega = 0.4$ and smaller spectral separation

(b) Lower frequency cutoff $\omega_{\max} = 2.5$ resulting in $\Delta\omega = 0.25$ and larger spectral separation

Figure 7: Spectral separation Δ_{sep} of the averaged TFIM cooling channel for $N = 4$, $\beta = 1$, $\sigma = 2$, for different coupling strengths α and frequency resolutions. The optimized configuration (NM) is given for comparison.

In Fig. 7 we can see that the spectral gap in general increases with the coupling strength α. Increasing the frequency resolution (by lowering $\omega_{\max}$) also increases the gap. However, $\omega_{\max}$ should not be smaller than the system's bandwidth. Interestingly the spectral gap is not uniform across the h/J-sweep but is reduced both in the highly ferro- and paramagnetic regimes.

For the considered parameter grid of 10 h/J values for $N = 4$ spins and different coupling strenghts α and temporal widths σ we computed the full spectra: here $\Delta_{\text{sep}} = \Delta_{\text{gap}}$ up to numerical precision. This shows that for for these parameters the spectral separation is a direct measure of slowest decay mode λ_2^n and thus is directly connected to the relaxation time towards the channel fixed-point. In particular, the calculation of the smallest finite eigenvalue $\mu_{\min}$ of the Lindbladian population transition matrix $(t)_{ij}$ is related to the eigenvalue $1 + \alpha^2\mu_{\min}$ of the restriction of $\mathcal{E}$ to classical mixtures $\sum p_i \left|i\right\rangle \left\langle i\right|$ in the H_0 eigenbasis. Noting that $\mu_{\min} < 0$ for decaying modes of $\mathcal{E}$, the closeness of Δ_{sep} and $\alpha^2|\mu_{\min}|$ is an indication that the Lindbladian approximation is accurate with respect to the slowest decaying mode near the fixed point. The further numerical inspection of the spectra indicates that the subleading eigenvalue λ_2 is positive and real, such that $\Delta_{\text{sep}} \approx \Delta_{\text{gap}}$ represents a sensible working hypothesis for our experimental runs.

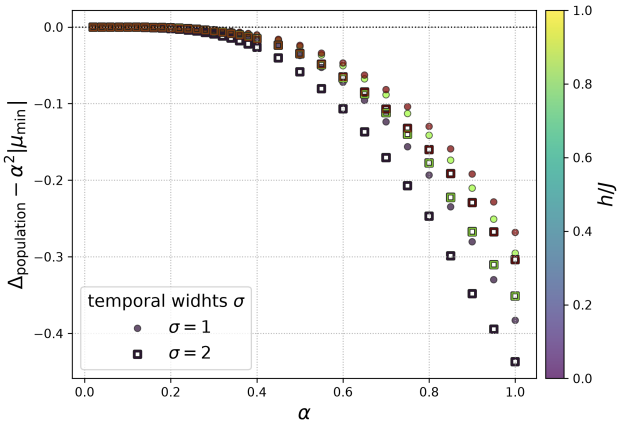


(a) Dependence of spectral gap of full channel $\mathcal{E}$.

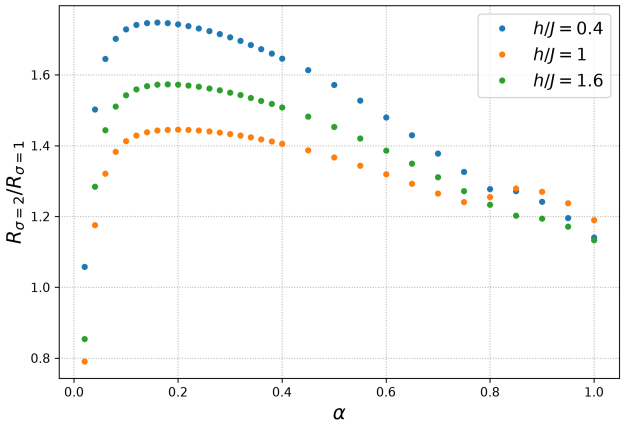
(b) Dependence of spectral gap of channel restricted to classical mixtures of $\rho = \sum_i p_i \left|i\right\rangle \left\langle i\right|$.

Figure 8: Ratio of channel gap Δ_{gap} and minimal distance $\mu_{\min}$ of nonzero eigenvector to zero. The Lindbladian evolution $\mathcal{E} \approx e^{\alpha^2 \mathcal{M}}$ predicts a quadratic dependence. Plots for two values of σ and three values of h/J (ferromagnetic 0.4, critical 1.0 and paramagnetic 1.6).

In Fig. 8 we compare the gap predicted by the generator via Lindbladian approximation both for the full channel $\mathcal{E}$ as well as for the restriction to the population sector. We find a good agreement up to medium coupling strengths of around $\alpha = 0.4$. For larger coupling strengths the deviation is considerable, but interestingly also differs qualitatively from a purely quadratic behavior insofar, as there are visible nonmonotonic effects.

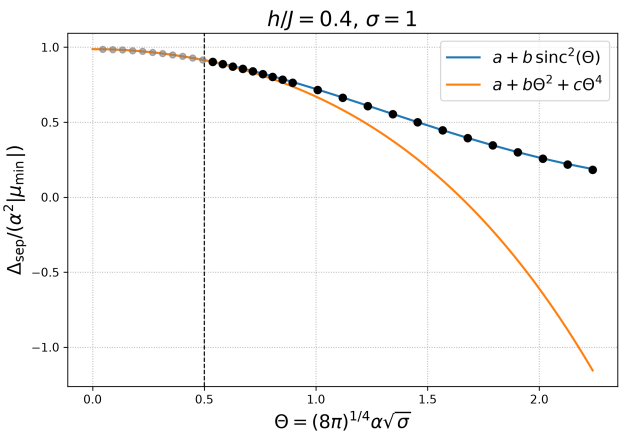


(a) Parameter dependence of signed residuals of the generator gap-estimate for $\sigma = 2$ and $\sigma = 1$.

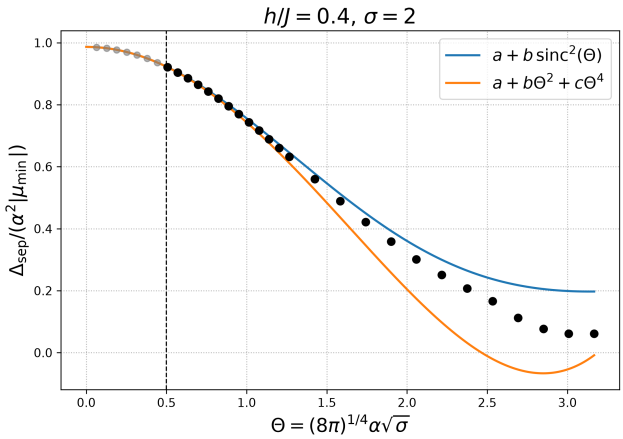


(b) Parameter dependence of residuals ratio from generator gap-estimate for $\sigma = 2$ and $\sigma = 1$.

Figure 9: Deviations of population-sector channel gap Δ_{gap} from the second-order α^2 generator approximation. Results for two values of σ and three values of h/J (ferromagnetic 0.4, critical 1.0 and paramagnetic 1.6) are shown.



(a) Approximation of the full channel spectral gap by the generator gap.



(b) Approximation of the population-sector spectral gap by the generator gap.

Figure 10: Dependence of the ratio of the actual estimated spectral gap on $\theta \propto \alpha\sqrt{\sigma}$. Comparison of the sinc^2 model and forth order polynomic model. Fitted for values $\theta < 0.5$ and validated for $\theta \geq 0.5$

In Fig. 9 we can better resolve the dependence on the temporal width and Hamiltonian parameter h/J. The ratio of the residues scales nonlinearly with α and also depends nontrivially on the regime h/J,, cf. Fig. 9b. However, when analyzing this behavior, finite size effects and coarse frequency resolution have also to be taken into account.

In Fig. 8 we can already see a nonmonotonic behavior. In Fig. 10 we extrapolate the data both with a sinc model and a standard fourth order polynomial expansion. A few comments are in

order. On resonance, i.e. for $E_j - E_i = -\omega$, the interaction Hamiltonian Eq. (22) is

$$H_I \approx \alpha \int_{-T}^{T} dt f(t) \left(\langle i|A|j\rangle \, |i,1\rangle \langle j,0| + \langle j|A|i\rangle \, |j,0\rangle \langle i,1| \right), \tag{52}$$

i.e. proportional to σ_x in the two-dimensional subspace spanned by the product states $|i,1\rangle$ and $|j,0\rangle$. For $T \gg 1$ holds approximately

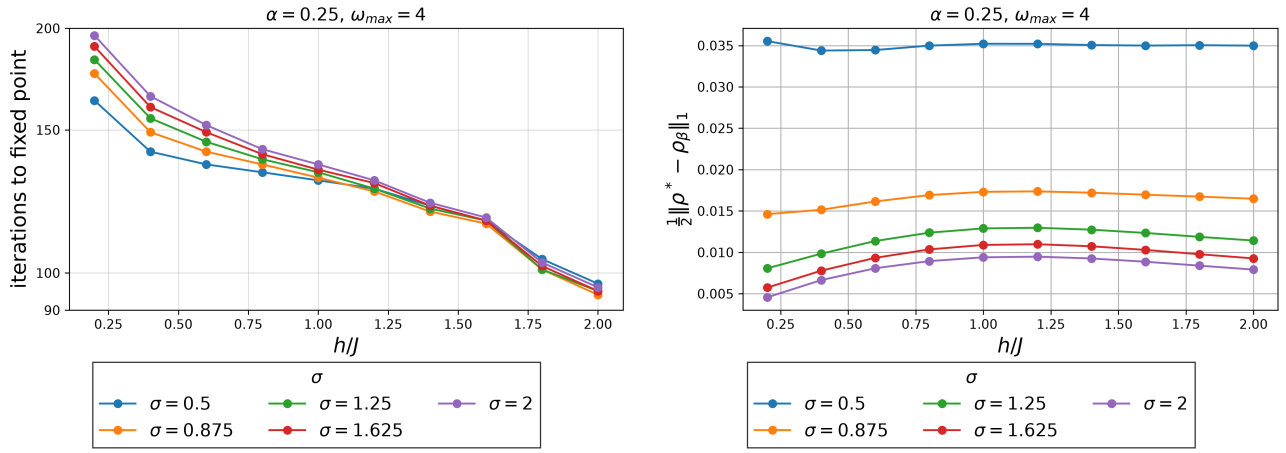
$$\theta := |\, \langle i|A|j\rangle \,| \cdot \alpha \sqrt[4]{8\pi\sigma^2} \propto \alpha\sqrt{\sigma}, \tag{53}$$

and The time evolution of the corresponding Hamiltonian describes a Rabi-oscillation with $p^{\mathcal{E}}_{i\leftarrow j} = \sin^2\theta$. Comparing this with the expression $p^{\mathcal{L}}_{i\leftarrow j}$ for the Lindbladian generator from Eq. (34), we get

$$\frac{p^{\mathcal{E}}_{i\leftarrow j}}{p^{\mathcal{L}}_{i\leftarrow j}} \approx \frac{\sin^2\theta}{\theta^2} = \mathrm{sinc}^2\,\theta. \tag{54}$$

Interestingly, a similar phenomenology is observed in the subspace of the two leading channel eigenstates also off resonance.

7.4 Filter width, temperature, and system-size dependence



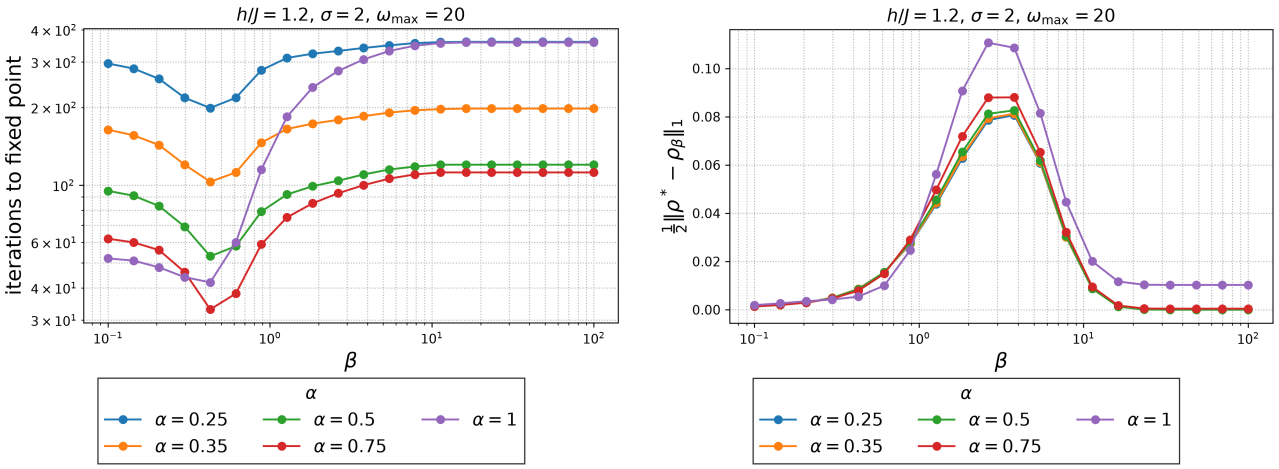
(a) Number of iterations.

(b) Trace distance to the Gibbs state.

Figure 11: Number of iterations required for convergence within trace-distance $\varepsilon = 0.05$ to the channel fixed point and trace distance between the channel fixed point and the Gibbs state for the TFIM cooling channel with $N = 4$, $\alpha = 0.25$, $\beta = 1$, and $\omega_{\max} = 4$, for different temporal widths σ.

We finally compare the number of iterations necessary to reach the channel fixed point within an $\epsilon = 0.05$ neighborhood as well as the trace distance to the Gibbs state for different temporal widths σ, see Fig. 11. We see that larger values of σ lead to a more accurate approximation

of the Gibbs state throughout the h/J-sweep. At the same time, the number of necessary iterations depends only weakly on σ, with larger widths requiring more iterations, in particular in the ferromagnetic regime.



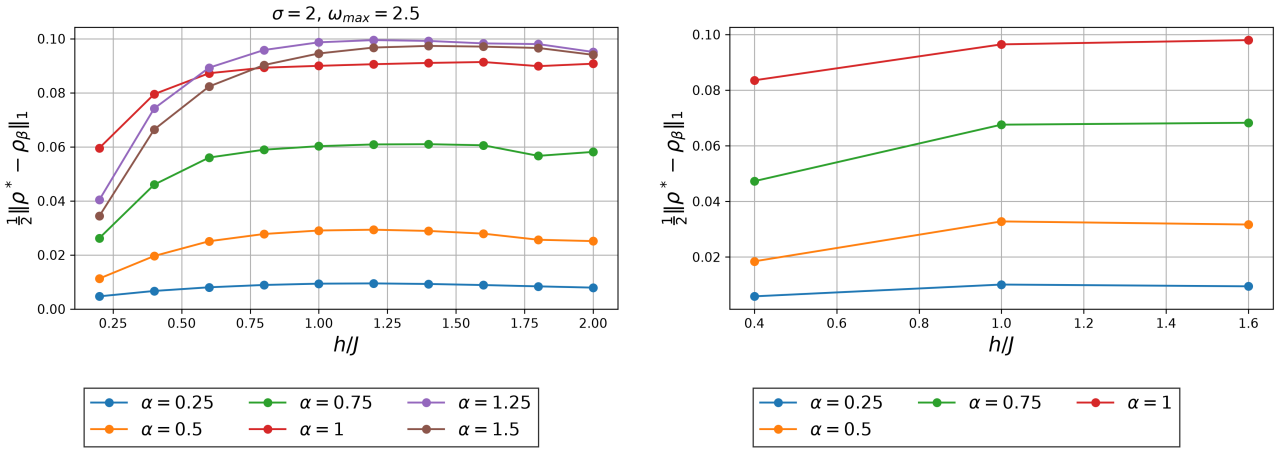
(a) Number of iterations.

(b) Trace distance to the Gibbs state.

Figure 12: Number of iterations required for convergence to the channel fixed point and trace distance between the channel fixed point and the Gibbs state for the TFIM cooling channel with $N = 4$, $J = 1$, $h = 1.2$, and $\sigma = 2$, for different coupling strengths α and inverse temperatures β. Here the Hamiltonian is *not* normalized according to Eq. (12).

In Fig. 12 we study the dependence on the inverse temperature β for different coupling strengths α. The number of necessary iterations first decreases, then increases up to a constant baseline for large β. The trace distance to the Gibbs state also exhibits a nonmonotonic dependence, with the largest deviations at intermediate β. Again, it is important to keep in mind that convergence to the channel fixed point does not necessarily imply closeness to the desired Gibbs state.

Finally, in Fig. 13 we compare the trace distance to the Gibbs state for $N = 4$ and $N = 6$. For both system sizes we observe the same qualitative behavior: smaller coupling strengths result in fixed points closer to the desired thermal state. For comparable coupling strengths and h/J values, the deviations are somewhat larger for $N = 6$. However to establish the scaling behavior with N more than two system sizes are needed.



(a) $N = 4$ spins.

(b) $N = 6$ spins.

Figure 13: Trace distance between the channel fixed point and the Gibbs state for the TFIM cooling channel with $N = 4$ and $N = 6$, $\beta = 1$, $\sigma = 2$, and $\omega_{\max} = 2.5$, for different coupling strengths α. Hamiltonians are normalized according to Eq. (12). The $N = 6$ sweep includes three h/J points at 0.4, 1.0, and 1.6.

8 Conclusions and Outlook

The main feature that can be concluded from the numerical experiments in this work is the trade-off between effective mixing time and correctness of the fixed point. This trade-off is most clearly controlled by the coupling strength α. Weak coupling gives the most accurate approximation of the Gibbs state, but requires many channel iterations. Increasing α accelerates convergence considerably, while at the same time sacrificing closeness of the fixed point to the desired thermal state. For the finite systems considered here, intermediate coupling strengths therefore provide the best compromise, specifically for values around $\alpha \approx 0.5 - 0.6$ clearly beyond weak coupling.

The spectral analysis of the associated quantum channels support these findings. In the weak-coupling regime the subleading channel eigenvalues are well described by a Lindbladian approximation. Deviations become increasingly important for larger values of α. The observed correction is nonlinear in α and also depends on σ and on the Hamiltonian regime. A sinc^2-model gives an accurate empirical description of the investigated parameter range and can be well motivated for an isolated resonant transition. However, no corresponding derivation for the spectral gap of the full averaged channel could be obtained.

The remaining parameters show a less pronounced but still relevant influence on the convergence. Increasing σ improves the fixed-point accuracy rather systematically, whereas the number of necessary iterations changes only moderately. The dependence on β is nonmonotonic and shows again that fast convergence to the channel fixed point should not be identified with a good

approximation of the Gibbs state. The limited comparison between $N = 4$ and $N = 6$ shows the same qualitative dependence on α, but does not allow for conclusions about the scaling with the system size.

There are several knobs that have not been turned in this work. Most prominently this concerns the set of coupling operators $\mathcal{A}$. The numerical experiments were restricted to $\mathcal{A}_{XZ} = \{X_i, Z_i\}$, although other sets such as $\mathcal{A}_{XYZ} = \{X_i, Y_i, Z_i\}$ and sets containing two-site operators have already been implemented. Since the coupling operators determine which transitions between energy eigenstates can be induced, it could prove worthwhile to investigate whether a different choice of $\mathcal{A}$ can improve the trade-off between mixing and fixed-point accuracy.

Another obvious direction concerns the frequency average resp. its numerical approximation. In the present implementation the bath frequencies are sampled from a uniform distribution and the averaging integral is replaced by a finite midpoint quadrature. A first step would be to systematically increase the frequency resolution and test convergence of the observed quantities. Beyond that, non-uniform frequency distributions or quadrature rules adapted to the relevant Bohr frequencies may be advantageous. The same applies to the choice of the filter function itself, which has not been varied apart from its width σ.

Further, the validation for larger spin chains is worth considering. The explicit construction of the full superoperator becomes infeasible quickly because its dimension grows as 4^N. But tensor-network methods, in particular MPS resp. MPO based simulations, could allow the channel dynamics to be studied for considerably larger systems. Even without access to the complete channel spectrum, convergence from selected initial states, thermal observables and the distance to an approximate target state could still be investigated. This would make it possible to test whether the parameter dependence found here survives beyond the very small system sizes accessible by exact diagonalization.

A more speculative direction would be to use a machine learning approach resp. a neural network for parameter optimization. Instead of separately optimizing α, σ, $\omega_{\max}$ and possibly properties of the frequency distribution for every Hamiltonian, one could try to learn suitable parameter choices from small training instances. Ideally, training on one particular class of Hamiltonians would generalize to related systems and, more importantly, to larger spin counts. It would be an interesting test of how universal the observed parameter dependencies actually are.

9 Repository

The code used for the numerical experiments is publicly available at `https://github.com/gabrielmaresch/groundstate_prep`.

The main implementation is contained in `cooling_channel.py`, which constructs the TFIM Hamiltonian and the system–bath cooling circuit, and in `superoperator.py`, which constructs the averaged channel and computes its fixed point and spectral properties. The files `ed.py` and `ed.jl` provide exact-diagonalization and Gibbs-state reference calculations, while `path_analysis.py` is used to study cooling trajectories. Parameter sweeps are performed using the `parallel_*.py` scripts, and the `nelder_mead_*.py` scripts implement the parameter optimization.

The corresponding analysis is organized in Jupyter notebooks: `grid_analysis.ipynb` and `sweep_analysis.ipynb` evaluate the parameter dependence, `generator_analysis.ipynb` compares the channel with the generator approximation, and `bohr_analysis.ipynb` examines the Bohr-frequency distribution. Numerical results are stored as `.npz` archives in `data/`, with separate subdirectories for grid calculations and optimization runs. The directory `docu/` contains this report and its figures in `docu/plots/`. Sample batch scripts for computation on the VSC are contained in `batch/` and `tests/` contains regression tests for the exact-diagonalization implementation. Installation instructions and examples for running the calculations locally are provided in `README.md`.

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